Preliminary results master thesis

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Abstract

The two MTU15 granularity reforms turn within-hour bidding into a strategic margin, and the dispatchable fleet exploits it. We see this directly in the bid curves: in critical hours (morning ramp and evening peak) where within-hour residual demand actually varies, ladders widen substantially — broadly under intraday reform (CCGT, hydro, pump-storage) and more selectively under the day-ahead reform (hydro, CCGT). Wind, whose output spans all hours rather than concentrating in peaks, shows little critical-flat differential.

Quantity-side, the cleared mix shifts in the direction the bid widening would predict: DA15 raises CCGT day-ahead auction-cleared volume by +59 GWh/day under joint 95% placebo-correction, and pump-storage redirects +237 MWh of its daily generation toward those same critical hours. Price-side, ID15 *cuts* clearing prices substantially in *both* IDA and DA markets — ~ -38 EUR/MWh raw and ~ -20 to -25 EUR/MWh net of a same-calendar 2024 placebo, conditional on daily wind generation, solar generation and gas prices (so the drop survives controlling for renewable supply). The cross-market symmetry (both markets drop by the same amount on the same day, even though ID15 only changed *intraday* granularity) is the signature of rational anticipation: traders compress the day-ahead risk premium once they know they can re-trade in finer 15-min slots intraday.

1. Setup: outcomes, windows, and seasonal control

> *Takeaway.* We measure how MTU15 reshapes three families of outcomes: per-curve bid shape; aggregate clearing price and cleared energy; and post-only within-hour dispersion. Identification compares *critical* clock-hours (those with the most within-hour load variation) to *flat* clock-hours (overnight; the within-day control) across each reform's pre/post window. Outcomes whose pre/post comparison crosses different seasons — chiefly DA15 prices and cleared quantities — are additionally seasonally adjusted.

(Two) reforms. The Spanish wholesale electricity market shifted from a 60-minute to a 15-minute market time unit in two steps: **ID15** (2025-03-19, intraday auctions and continuous market) and **DA15** (2025-10-01, day-ahead). After each reform there are four periods per clock-hour; before, one. The *operación reforzada* regulatory stance (REE's post-blackout reinforced operation, continuous from 2025-04-28) is a continuous confound that forces CCGT post-dayahead inclusion; tight pre/post windows around each reform date hold its level constant within each arm. Figure 1 summarises the full timeline, including other reforms that are not directly relevant for our research question but are important for context.

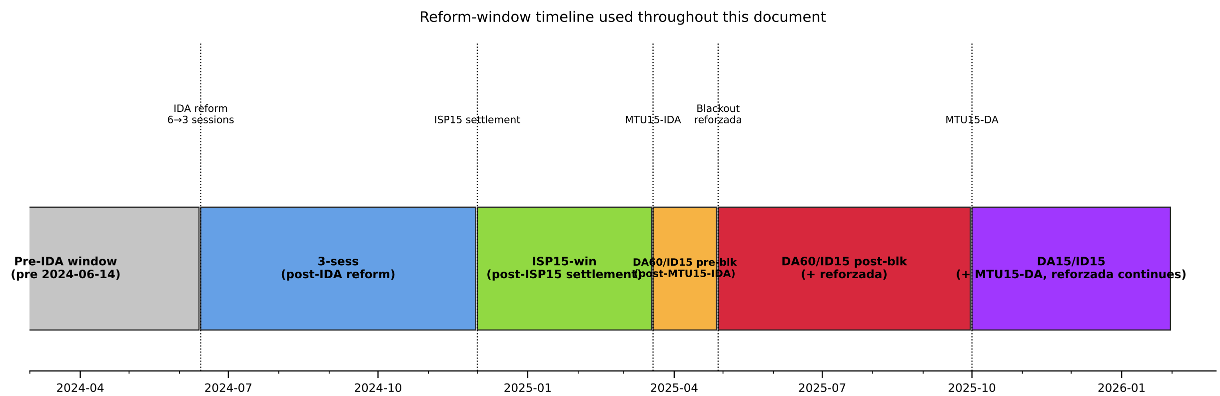

Figure 1: **Full reform-window timeline.** Five named windows partition the post-2024-06-14 sample; the pre-IDA window is the baseline regime. Dotted vertical lines mark the five reform dates that produce the window boundaries.

Variables. Each (unit u, date d, period p) sell offer is a tranche ladder $\{(p_k, q_k)\}_{k=1}^{K_{u,d,p}}$ where p_k is the bid price (EUR/MWh) and q_k the quantity in MW[1]. Offers carry one of three structural types: *simple* (a bare price–quantity ladder), *minimum-income condition* (MIC), or *block* (all-or-nothing across periods). Most technologies use simple bids, but CCGT is the outlier with the bulk of its offers in MIC or block form (§ 4, § 5(d))[2]. Let $\mathrm{MCP}_{d,p}$ be the OMIE clearing price *of the market in which the offer was submitted* and h a bandwidth that isolates the bids in contention: tranches close enough to the clearing price to plausibly compete in the *competing tier* of the offer (CCGT's two-tier structure makes this distinction explicit, § 7). We anchor h to the *contemporary contested margin*: h is computed as $p_{90} - p_{50}$ of the MCP distribution *in the analysis window of the comparison, market by market*. The eight bandwidths used in this memo (one per reform $\times$ {real, placebo} $\times$ {DA, IDA}) are: ID15 real DA = 50, ID15 real IDA = 62, ID15 placebo DA = 45, ID15 placebo IDA = 46, DA15 real DA = 50, DA15 real IDA = 58, DA15 placebo DA = 45, DA15 placebo IDA = 49 EUR/MWh. Bandwidth-robustness across wider h choices is reported in § 5(e). Now define the in-band set, MW weights, and in-band weighted mean bid price:

$$\mathcal{B}_{u,d,p} \;=\; \{k : |p_k - \mathrm{MCP}_{d,p}| \le h\}, \qquad w_k \;=\; \frac{q_k}{\sum_{j\in\mathcal{B}_{u,d,p}} q_j}, \qquad \bar{p}_{u,d,p} \;=\; \sum_{k\in\mathcal{B}} w_k\, p_k.$$

Per-curve (in-band, MW-weighted; one value per (unit, date, period)):

$$\sigma_{p,\,u,d,p} \;=\; \sqrt{\sum_{k\in\mathcal{B}_{u,d,p}} w_k\,(p_k - \bar{p}_{u,d,p})^2}, \qquad N_{\mathrm{eff},\,u,d,p} \;=\; \Big(\sum_{k\in\mathcal{B}_{u,d,p}} w_k^2\Big)^{-1}.$$

σ_p = price spread of the bid ladder within a single quarter; N_{eff} = effective tranche count of that quarter (1 = a single block, large = many equal-sized tranches).

Aggregate: $\mathrm{MCP}_{d,p}$ is the OMIE clearing price (EUR/MWh); $Q_{d,p} = \sum_{u,k} q^{\mathrm{cleared}}_{u,d,p,k}$ is the system DA cleared quantity per (date, period)[3].

[1] Following OMIE *Potencia* field in *ficheros OMIE v1.37* §5.1.4.2.

[2] Approximate sell-offer shares by structural type (DA15/ID15): wind, solar, hydro and pump-storage ∼100% simple; nuclear 64% simple, 36% MIC; CCGT 18% simple, 50% MIC, 32% block.

[3] The OMIE assigned-power field is per-period instantaneous power (MW), invariant to period length, so one 15-min

Within-hour dispersion (post-only, used to calibrate the critical/flat partition in § 2): let $m_q = \sum_{k \in \mathcal{B}_q} q_k$ and $\bar{m} = \frac{1}{4} \sum_q m_q$. For each (unit, date, clock-hour) we compute the standard deviation across the four quarters of the clock-hour for each of the five outcomes[4]:

$$D_{\text{price}} = \text{sd}_q(\bar{p}_q), \quad D_{\text{qty}} = \text{sd}_q(m_q)/\overline{m}, \quad \text{SD}_{\text{price}} = \text{sd}_q(\text{MCP}_q),$$
$$D_\sigma = \text{sd}_q(\sigma_{p,q}), \quad D_N = \text{sd}_q(N_{\text{eff},q}).$$

D_{price}, D_{qty}, SD_{price} are across-quarter shifts in bid LEVEL, bid VOLUME, and system clearing prices; D_σ, D_N are across-quarter shifts in ladder SPREAD and TRANCHE COUNT. All five are zero under the MTU60-equivalent strategy (identical bids in every quarter), so any positive value post-MTU15 is a direct trace of within-hour discrimination.

Method and seasonal control. Bid shape uses a DiD with the critical/flat partition (per-curve, ∼3-month windows around each cutover); clearing prices and per-tech cleared energy use a Bayesian Structural Time Series counterfactual (Brodersen et al., 2015, applied to the MTU15 EPEX reform by Märkle-Hußet al. 2020) on daily series with wind+solar+gas covariates, regime-constant pre-windows (DA15 starts after reforzada onset 2025-04-28; ID15 ends before the blackout), and a 2024 same-calendar placebo for residual cross-season trend (full windows in § 3). Bid-shape outcomes need no extra seasonal control: σ_p and N_{eff} are MCP-centered and MW-weighted on the in-band ladder, so price-level seasonality does not transmit; the flat-hour control absorbs common-to-all-hours bid-shape seasonality and the pre-trend placebo (§ 5(b)) checks the critical−flat differential itself. The supporting pump-storage DiD uses a same-calendar-month restriction (§ 5).

2. The critical/flat partition: where granular pricing has economic content

Takeaway. Within-hour discrimination has economic content only where within-hour residual demand actually varies. Critical hours (morning ramp + evening peak) are where MTU15 can bite; flat hours (overnight) are the within-day control. Post-MTU15 within-hour dispersion is much larger in critical than flat hours on every bid-level and bid-shape outcome we measure — the partition correctly identifies where the reform lives.

Intuition. Granular pricing only has economic content where within-hour residual demand[5] actually varies; if a clock-hour is internally flat, the four quarters look the same and there is nothing to discriminate on. We measure within-hour variation by $\sigma_{\text{within}}(h, d) = \text{sd}_q[\text{load} - \text{wind} - \text{solar}]_{q,h,d}$ on 15-min residual-demand data[6] and partition clock-hours by median σ_{within}:

quarter and one 60-min hour at the same MW are directly comparable as rates. Aggregating MW across the four quarters of a clock-hour gives the stock of energy delivered in that clock-hour, $\text{MWh}_h = \sum_{q \in h} \text{MW}_q \, \Delta t_q$. Aggregate-quantity coefficients below are reported as MWh per clock-hour (numerically equal to the per-period mean MW because the integration is over 1 h, but the unit is energy, not power).

[4] We ommit the u, h subscripts for readability.

[5] Spanish-market convention calls this the *hueco térmico* (thermal gap) — the load that thermal and other dispatchable generation must cover after subtracting wind and solar.

[6] ENTSO-E Transparency Platform: A65 (actual total load) and the wind/solar component of A75 (actual generation per type, PSR codes B16, B18, B19); window 2023-01-01 to 2026-05-14.

- **Critical** $\mathcal{C} = \{5, \ldots, 8,\ 16, \ldots, 22\}$: morning ramp + evening peak; $\sigma_{\text{within}} \in [522, 1303]$ MW.
- **Flat** $\mathcal{F} = \{1, 2, 3\}$: overnight; $\sigma_{\text{within}} \in [336, 357]$ MW.
- **Midday** $\{11, \ldots, 14\}$ (solar-marginal, $\sigma_{\text{within}} \in [468, 502]$ MW): excluded from main specs; kept for falsification (§ 5(a)).

Formal calibration of the partition (post-only). Let $D_{d,h}$ stand for any one of the five within-hour dispersion outcomes (D_{price}, D_{qty}, SD_{price}, D_σ, D_N) on date d, clock-hour h. Under MTU60 the within-hour quarter dimension does not exist, so $D_{d,h}$ is mechanically zero pre-reform; we use the post-MTU15 sample to *describe* where within-hour dispersion lives across hour-classes. A date-FE regression is a convenient way to summarise the critical−flat differential:

$$D_{d,h} \;=\; \delta_d \;+\; \theta\,\mathbf{1}\{h \in \mathcal{C}\} \;+\; \varepsilon,$$

fit per outcome. θ is the post-MTU15 critical−flat dispersion gap. This is a **descriptive contrast, not a causal estimate**.

Outcome	**Subsample**	**ID15**	**DA15**	**crit/flat ratio (IDA, DA)**
D_{price} (EUR/MWh)	per unit (pooled)	0.39^{***} (0.05)	1.35^{***} (0.11)	4.1×, 7.4×
D_{qty} (CV)	per unit (pooled)	0.024^{***} (0.003)	0.033^{***} (0.002)	4.3×, 4.9×
SD_{price} (EUR/MWh)	system	4.55^{***} (0.40)	3.97^{***} (0.37)	2.8×, 2.6×
D_σ (EUR/MWh)	CCGT	−0.24 (0.35) ns	$+0.57^{***}$ (0.10)	ns, 5.9×
	Hydro	$+0.48^{***}$ (0.07)	$+1.22^{***}$ (0.16)	2.2×, 3.1×
	Hydro pump	$+0.16^{***}$ (0.03)	$+0.66^{***}$ (0.15)	3.7×, 10.6×
	Wind	−0.003 (0.005) ns	$+0.04^{***}$ (0.02)	1.0×, 2.3×

SE clustered by date; $^{***}p < 0.01$, ns = not significant. Bid-level rows (D_{price}, D_{qty}, SD_{price}) are bandwidth-free. The per-tech D_σ rows are computed at the new p_{90} baseline (§ 1): ID15 column uses post-ID15 IDA data with h=62; DA15 column uses post-DA15 DA data with h=50. *Reading.* Under DA15, D_σ is positive and significant for all three dispatchable techs in DA — CCGT +0.57, hydro +1.22, pump-storage +0.66 — with the largest critical/flat ratio for CCGT (5.9×). Under ID15, hydro and pump-storage widen their IDA within-hour ladders (+0.48 and +0.16), while CCGT IDA D_σ is not distinguishable from zero — consistent with combined-cycle gas mostly competing in DA rather than IDA at this stage. Wind stays small everywhere, the price-taking baseline.

3. Identification: DiD for bid shape, BSTS for prices and quantities

Takeaway. Bid shape is identified by a critical/flat DiD: the (critical − flat) shift across each cutover is the ATT under parallel trends. Clearing prices and per-tech cleared energy are identified by a Bayesian Structural Time Series counterfactual on a daily panel: conditional on the wind and solar renewable-supply covariates, the post-cutover series departs from the model's pre-trained extrapolation by the treatment effect; a 2024 same-calendar placebo gauges residual cross-season trend.

u unit; d date; p period; h clock-hour; s IDA session; T_r reform date. Treatment status $D_{d,h} \equiv \mathbf{1}\{d \geq T_r\} \cdot \mathbf{1}\{h \in \mathcal{C}\}$; sample restricted to $h \in \mathcal{C} \cup \mathcal{F}$ for the DiD.

3.A. Spec A: Bayesian Structural Time Series for prices and per-tech cleared energy

Following Brodersen et al. (2015) as applied to the EPEX 15-min reform by Märkle-Hußet al. (2020), the daily response y_t (DA or IDA clearing price; per-tech daily cleared GWh) is modelled as a Gaussian state-space $y_t = \mu_t + \gamma_t + X_t'\beta + \varepsilon_t$, with μ_t a local-level random walk with stochastic slope, γ_t a 7-day weekly cycle, and $X_t = (\text{wind GWh}, \text{solar GWh}, \text{gas EUR})$ exogenous covariates under a spike-and-slab prior on β (gas added to Märkle-Hußet al.'s wind+solar because Spain's gas exposure differs from Germany's; demand excluded to avoid simultaneity). Pre-cutover data train the model in `CausalImpact`; the counterfactual $\hat{y}_t(0)$ on the post-window is the extrapolation under realised X_t, and $\bar{\tau} = \frac{1}{|\text{post}|}\sum_{t \geq T_r}(y_t - \hat{y}_t(0))$ with its 95% credible interval is read from the MCMC draws. Cleared-quantity outcomes are reported in absolute GWh/day rather than % (the Gaussian state-space can produce negative posterior draws when covariates predict very low CCGT dispatch, and %-readings would re-introduce log-like unit-dependence issues, Chen and Roth 2024).

Windows hold the relevant regulatory regime constant across pre and post. DA15: pre 2025-04-28 → 2025-09-30 (156 d, starting at reforzada onset); post 2025-10-01 → 2025-11-09. ID15: pre 2024-06-14 → 2025-03-18 (278 d, post-IDA-reform 3-session architecture); post 2025-03-19 → 2025-04-27 (40 d, pre-blackout). A 2024 same-calendar placebo for each reform falsifies residual cross-season trend; placebo-net effects are the cleanest identification of the treatment. *Identifying restriction:* conditional on X_t, $y_t(0)$ post-cutover equals the pre-trained extrapolation; the placebo gauges whether that extrapolation is itself biased by seasonality.

3.B. Spec B: per-hour-class BSTS on bid-level outcomes

Spec C below identifies the within-quarter shift in bid *shape*. To identify within-day reallocation in bid *levels*, we run Spec A-style BSTS *independently* on each (tech, market, hour-class) cell for two bid-level outcomes: $\bar{p}_{t,m,h,d}$ (MW-weighted mean of in-band tranche prices) and $m_{q;t,m,h,d}$ (total in-band quantity, summed across tranches/periods/units and for IDA also sessions). Hour-classes follow the calibration of § 2; same wind+solar+gas covariates and same windows as § 3.A; the in-band bandwidth is window-and-market-specific $h = p_{90} - p_{50}$ of MCP (eight values in § 1). Within each (tech, market, outcome) the identified target is the (critical − flat) placebo-net contrast, post-processed from four BSTS posteriors (real/placebo × critical/flat); the joint 95% credible interval treats the four posteriors as independent. We run BSTS per hour-class and form the contrast in post-processing (rather than apply BSTS to the differenced series) so the seasonal structure stays available for the model to absorb.

3.C. Spec C: per-curve DiD for bid shape

For $Y \in \{\sigma_p, N_{\text{eff}}\}$, per tech:

$$Y_{u,d,p[,s]} = \alpha_u + \beta\,\mathbf{1}\{d \geq T_r\} + \xi\,\mathbf{1}\{h \in \mathcal{C}\} + \theta\,D_{d,h} + \varepsilon, \qquad \text{SEs clustered by date.}$$

We use DiD rather than BSTS here because the bid-shape outcomes are MCP-centered and MW-weighted (§ 1), so they do not inherit price-level seasonality. Differencing absorbs any common-hour seasonality, and under parallel trends on the (critical − flat) differential, θ identifies the ATT on critical-hour bid shape. Flat hours act as a low-exposure within-day control (mechanically reachable post-cutover but with negligible within-hour residual demand variance, § 2), giving the standard 2×2 DiD without staggered-treatment negative weights (de Chaisemartin and D'Haultfœ uille, 2026, ch. 3).

4. Results: bid-shape widening, day-ahead CCGT auction-cleared scale-up, and directional price movements

We report findings one spec at a time, A then B then C, mirroring the order of § 3: Spec A (Bayesian counterfactual on daily prices and per-tech cleared energy, § 3.A), Spec B (per-hour-class BSTS on bid-level outcomes — mean in-band price and total in-band quantity, § 3.B), and Spec C (per-curve DiD on bid shape, § 3.C). Spec A identifies daily-level effects on clearing prices and per-tech cleared energy; Spec B identifies within-day reallocation of bid *levels* across hour-classes; Spec C identifies within-quarter bid-*shape* effects. All three share the same wind+solar+gas covariates and the 2024 same-calendar placebo. An integrative takeaway closes the section.

4.A. Spec A: prices and per-tech cleared energy

		ID15		**DA15**	
Outcome	**Tech**	Effect	Placebo	Effect	Placebo
Clearing price (EUR/MWh)	—	-38.34^{***}	-18.08^{*}	$+2.97$	-8.54
Cleared (GWh/day)	CCGT	-8.08^{***}	-9.18^{***}	$+40.31^{***}$	-18.69^{***}
	Hydro	-1.27^{***}	$+4.98^{***}$	$+6.16^{*}$	$+9.28^{***}$
	Hydro pump	-0.37^{*}	$+0.54^{**}$	$+4.43^{***}$	$+1.42^{*}$
	Wind	$+5.34^{***}$	$+7.99^{***}$	$+13.16^{***}$	$+2.15$
	Solar	$+6.70^{***}$	$+2.55^{***}$	-3.37	$+1.38$
	Nuclear	$+11.51^{***}$	$+19.06^{***}$	-18.21^{**}	-34.77^{***}

Bayesian posterior point estimates; tail-probability stars $^{*} < 0.10$, $^{**} < 0.05$, $^{***} < 0.01$. Windows in § 3.A; placebo columns are the same calendar window in 2024 with a fake cutover. The placebo-net effect (Effect − Placebo) is the identified treatment; the joint 95% credible interval propagates both posterior variances. Observed-vs-counterfactual series in Figures 2–5.

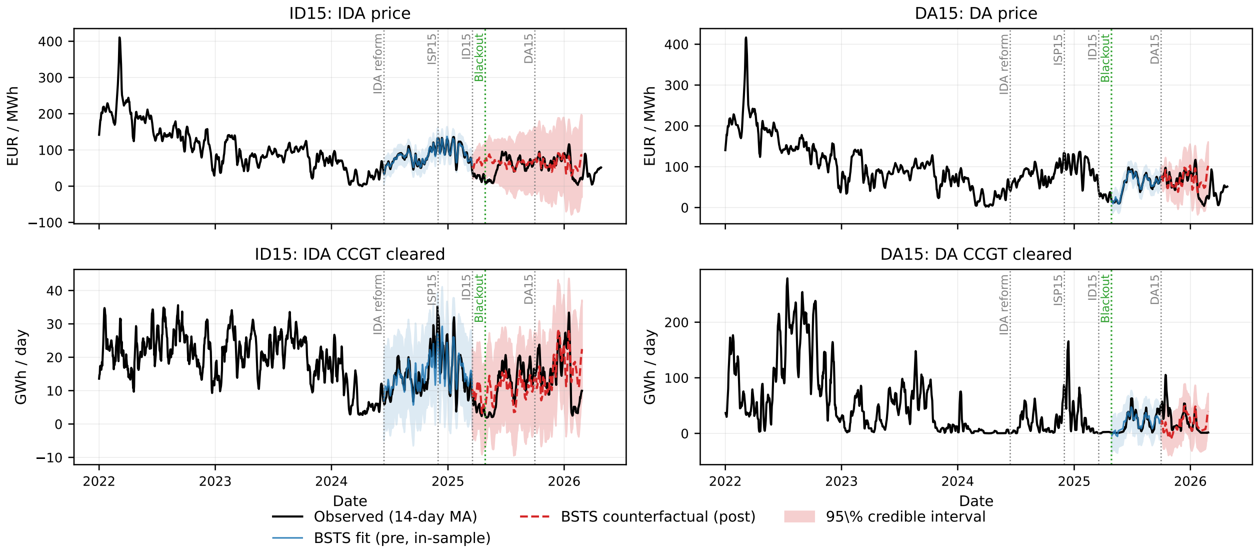

Figure 2: **BSTS counterfactual against the full 2022–2026 panel**, four headline outcomes (IDA price, DA price, IDA CCGT cleared, DA CCGT cleared; 7-day moving average). Black: observed. Blue: in-sample fit on the pre-window with 95% band. Red dashed: out-of-sample counterfactual with 95% band. Vertical lines mark the five reform/event dates (IDA reform, ISP15, ID15, blackout in green, DA15). The blue fit hugs the observed series across the full pre-window — wind+solar+gas covariates plus the local-linear trend absorb most of the variation, so the post-cutover gap is the treatment signal.

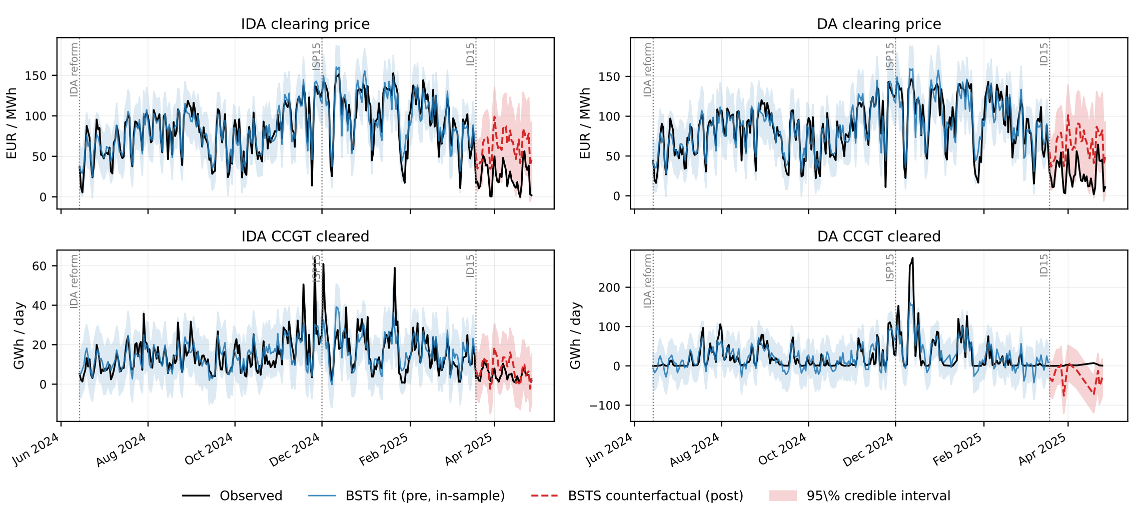

Figure 3: **ID15 effects on both markets** (2×2 zoomed view, 40-day pre-blackout post-window). Same colour scheme as Figure 2. Cutover 2025-03-19; blackout 2025-04-28. Both IDA and DA clearing prices (top row) drop below the counterfactual by similar magnitudes on the same day, even though ID15 only changed intraday granularity — the cross-market rational-anticipation channel into day-ahead (finding (ii)).

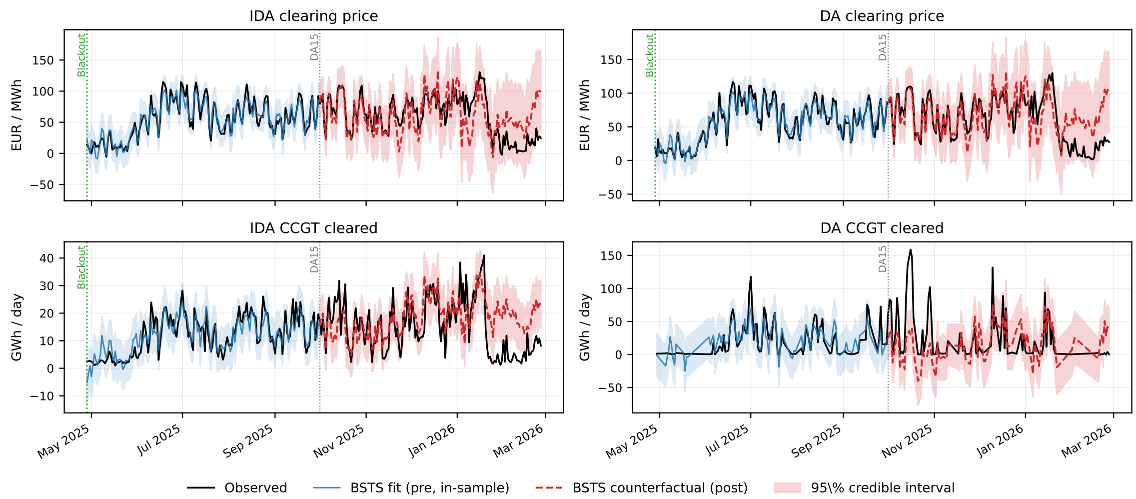

Figure 4: **DA15 effects on both markets** (2×2 zoomed view, post-window through 2026-02-26). Same colour scheme as Figure 2. Cutover 2025-10-01; reforzada-constant pre-window starts 2025-04-28. The DA CCGT panel (bottom-right) is the one quantity outcome where the observed series visibly departs from the counterfactual post-cutover — the surviving CCGT scale-up of finding (iii). By contrast, the IDA panels show no backwards transmission (intraday clearing follows day-ahead, so a DA-side reform cannot propagate backwards).

(iii) [Surviving] Day-ahead combined-cycle gas auction-cleared volume scales up under DA15. The headline magnitude is in the BSTS table; this is the *only* level-shift finding that survives the conservative joint 95% test net of the 2024 same-calendar placebo. The channel is the same as the bid-shape widening in (i): MTU15-DA lets combined-cycle gas discriminate finely within the hour, raising its expected acceptance probability across the four quarters of a peak hour. Extending the post-window through the full available winter shrinks the estimate (Figure 4, bottom-right) — still positive, but pulled down as observed cleared volume mean-reverts after a strong October. Hydro, solar and nuclear are essentially null net of the 2024 placebo. **[NOVEL]** The scale-up is essentially *Naturgy's*: per-firm-class daily DA cleared GWh placebo-net (post − pre, real − 2024 placebo) is GN +7.94, IB −2.37, GE −1.01, HC −1.18, fringe −0.77. The same firm that drives the DA15 CCGT σ_p widening in § 5 drives the cleared-MW scale-up. **[NOVEL]** The CCGT cleared scale-up is one-sided: +30.5 GWh/day placebo-net on DA but only +0.5 on IDA — combined-cycle gas migrates *into* DA without losing the IDA arm in cleared terms, so the bid-side ladder migration in (v) does not translate symmetrically to clearing.

(ii) [Wounded] ID15 raw clearing-price drop in *both* markets, but the conservative joint test does not retain. The raw drop is the cleanest reform-attributable signal in the price data (BSTS table; top row of Figure 3). Three credibility signals stack: the BSTS already conditions on wind, solar and gas, so the drop is what remains after the merit-order shift; both markets drop by the same magnitude on the same day even though ID15 only changed *intraday* granularity, the rational-anticipation channel into day-ahead; and across the pre-window the BSTS hugs the observed series tightly (Figure 3). The placebo-net joint 95% credible intervals do bracket zero, but this is a conservative test that propagates two posterior variances; the placebo's own loose CI reflects recurring spring-month model residual rather than a clean null. The DA15 price effect, by contrast, is small and not directional. *Verdict.* The finding is wounded by the 2024 same-calendar placebo (§ 5(c)): the placebo-net joint test does not retain it, so we report the raw drop and its rational-anticipation signature without claiming a surviving level effect. **[NOVEL]** The rational-anticipation channel also shows up in *cleared quantities*: CCGT DA cleared rises +25.6 GWh/day placebo-net under ID15 (the IDA-side reform) — an IDA-only granularity instrument generates a same-day DA-side CCGT scale-up, mirroring the DA σ_p widening in (i).

(iv) Pump-storage daily cleared MW: ID15 intraday-auction decrease surviving; DA15 day-ahead rise borderline. Figure 5 shows both effects against their 2024 placebos. The tech with the largest within-day arbitrage value reallocates its dispatch once each market becomes 15-min-priceable. ID15 pulls pump-storage capacity *out of* the now-finer intraday auction (top-left below the band, top-right placebo flat; placebo-net survives joint 95%). DA15 lets the day-ahead auction absorb more of its discharge (bottom-left above the band) — the closed-form storage prediction (§ 3) — but the placebo (bottom-right) also rises, so the daily-level test is borderline; the within-day redistribution shows up cleanly in the bid-shape geometry of (i) and the per-hour-class quantity migration of (v) instead.

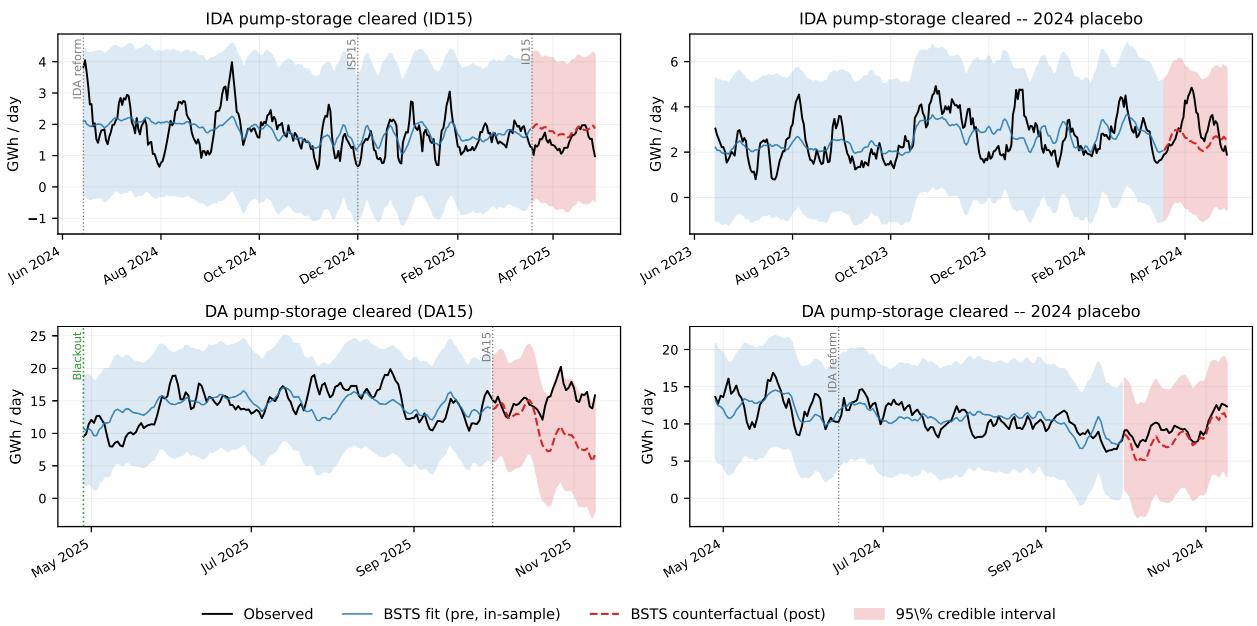

Figure 5: **BSTS counterfactual for pump-storage cleared MW.** *Top row:* ID15 effect on intraday-auction pump-storage cleared (left) and the 2024 same-calendar placebo (right). *Bottom row:* DA15 effect on day-ahead pump-storage cleared (left) and its 2024 placebo (right). Colour scheme as Figure 2.

4.B. Spec B: within-day reallocation under MTU15 (per-hour-class BSTS)

Figure 6 reports the *placebo-net level* of the BSTS effect for each hour-class (critical, midday, flat) separately, rather than just the (critical − flat) differential. The differential collapses information — two hour-classes can move in the same direction at different magnitudes, and the differential then hides the level story. The level view also makes explicit a caveat: these outcomes are *bid-level* ($\bar{p}$

and $\sum m_q$ are computed on submitted bids in the window-and-market-specific $\pm h$ EUR/MWh band around MCP, with $h \in \{45, 46, 49, 50, 58, 62\}$ as set in § 1), not cleared/dispatched quantities. Cleared quantity is captured by Spec A in § 4.A; the two can move in opposite directions (e.g., under ID15 CCGT's BSTS daily cleared MW falls while its in-band offered MW rises, because more tranches qualify as "in-band" when MCP itself falls).

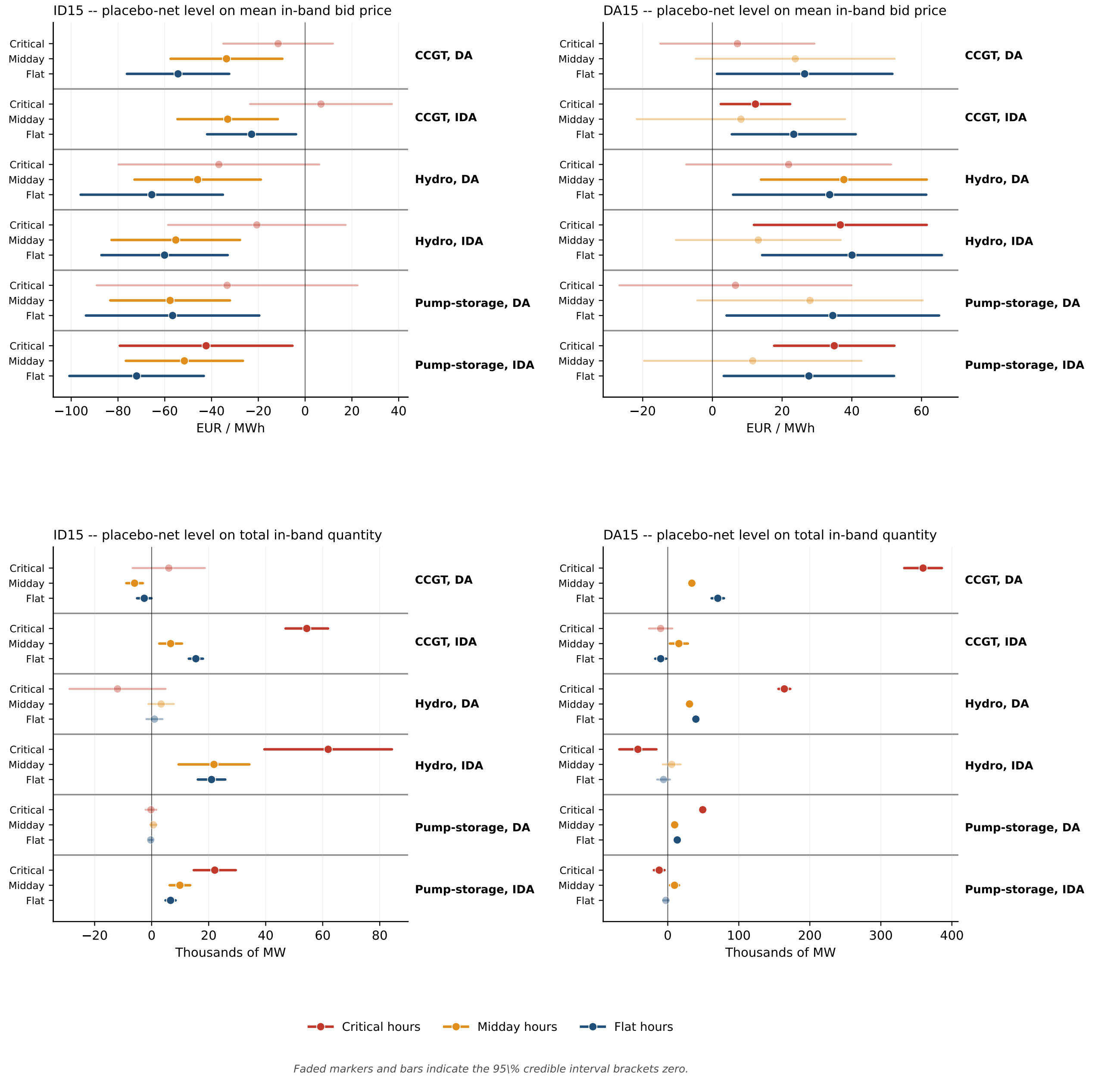

Figure 6: **Per-hour-class BSTS on bid-level outcomes — placebo-net levels.** Each row is one BSTS run. The (tech, market) group is bold-labelled on the right; within each group, three rows give the placebo-net effect for critical (red), midday (orange) and flat (blue) hours separately. Error bars are the 95% credible interval propagating real and placebo posterior variances. *Faded markers and bars* indicate the 95% CI brackets zero (effect not statistically distinguishable from zero). *Top row of panels:* mean in-band bid price $\bar{p}$ (EUR/MWh). *Bottom row of panels:* total in-band quantity $\sum m_q$ (thousands of MW, summed across tranches, periods, units, and — for IDA — sessions). *Left column:* ID15. *Right column:* DA15. *Caveat:* these are *submitted-bid* outcomes, not cleared/dispatched MW; the in-band test (window-and-market-specific $\pm h$, $h \in \{45, 46, 49, 50, 58, 62\}$; see § 1) is MCP-relative, so a drop in MCP can mechanically pull more tranches inside the band without any change in bidding behaviour.

(v) [Surviving] Under ID15 the dispatchable fleet migrates in-band offering *into* IDA across hour-classes and bid prices fall; under DA15 the same fleet migrates from IDA *into* DA across hour-classes and bid prices rise. (Magnitudes in Figure 6; surviving cells are full-colour markers.) The substitution picture is sharp on both outcomes:

- *In-band quantity:* under ID15, in-band MW grows in IDA across all three hour-classes for combined-cycle gas, hydro and pump-storage (peaks: hydro IDA critical +62,000 MW; CCGT IDA critical +54,000); under DA15, in-band MW grows in DA across hour-classes (CCGT DA critical +359,000 MW) and falls in IDA. The shift is comprehensive across the day, not concentrated in critical hours — dispatchables migrate the *entire competing-tier ladder* into the market that has just become granular.
- *Bid-price level:* ID15 drives placebo-net mean in-band bid prices down by –20 to –70 EUR/MWh across cells (sharper in flat/midday); DA15 raises them by +10 to +40 across cells. The placebo-net contrast strips the calendar trend; the surviving effects identify firm-side restructuring of competing-tier bid levels.

Bid vs. clearing caveat. Granularity in market m attracts in-band *bidding* into m (the upstream-migration channel the model in § 3 predicts). Whether the migration translates to cleared MW is a separate question answered by Spec A: DA15 CCGT cleared MW does scale up (finding (iii)); ID15 CCGT cleared MW *drops* even as in-band offered MW expands — the same MCP-relative artefact that lets more tranches fall "in-band" when MCP itself falls.

Tech selection for Spec C. CCGT (the focal scarcity-tier technology), hydro and hydro pump for the dispatchable comparison, and wind as a price-taking falsifier.[7] Solar PV is excluded (zero in-band scarcity-tier capacity); nuclear is excluded (mostly bilateral). Hydro, hydro pump and wind are ∼100% simple offers in both arms, so their Spec C DiDs are simple-only by construction; the offer-type composition concern is CCGT-specific.

4.C. Spec C: bid-shape widening in critical hours

		ID15		**DA15**	
Outcome (DiD θ)	**Subsample**	DA (h=50)	IDA (h=62)	DA (h=50)	IDA (h=58)
σ_p (EUR/MWh)	CCGT	$+4.17^{***}$ (0.44)	+0.10 (0.53)	$+0.68^{***}$ (0.22)	-0.68^{*} (0.36)
	Hydro	-0.53^{*} (0.29)	$+0.43^{*}$ (0.25)	$+1.80^{***}$ (0.34)	+0.30 (0.25)
	Hydro pump	-0.73^{***} (0.27)	−0.01 (0.18)	$+0.92^{***}$ (0.35)	$+0.32^{*}$ (0.19)
	Wind†	-0.10^{***} (0.03)	$+0.04^{**}$ (0.02)	−0.02 (0.03)	-0.02^{***} (0.003)
N_{eff}	CCGT	−0.08 (0.21)	+0.79 (0.73)	$+1.89^{***}$ (0.15)	−0.17 (0.53)
	Hydro	-0.24^{***} (0.06)	$+0.18^{***}$ (0.05)	$+0.51^{***}$ (0.09)	+0.00 (0.06)
	Hydro pump	+0.07 (0.05)	$+0.12^{*}$ (0.07)	−0.06 (0.07)	−0.16 (0.13)
	Wind†	+0.01 (0.02)	$+0.03^{***}$ (0.01)	−0.01 (0.01)	−0.01 (0.01)

[7] Wind has the dominant sell-side quantity but is not the marginal price-setter; CNMC (2025c) defines the marginal technologies as coal + CCGT + hydro + 5% RECORE. Aggregator portfolios construct dense single-price blocks at low EUR/MWh, so the clearing price lands inside a block and EUPHEMIA partially accepts a wind unit as the mechanical residual — not strategic price-setting. A strategically inert technology should yield small DiDs, validating the contrast.

Notes: SEs in parentheses, clustered by date. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$. Per-(tech, market) subsample regressions, unit FE absorbed via within-unit demeaning. Bandwidth h is window-and-market-specific (the p90 of the MCP distribution in that window, market by market): see § 1 and § 3.C. Sensitivity to wider bandwidths is reported in § 5(e). †Wind reads as a price-taking baseline: 84% of in-band wind sell curves are a single block; magnitudes are an order of magnitude smaller than the dispatchable techs.

(i) Bid-shape widening in critical hours is market-specific: ID15 widens day-ahead CCGT and intraday N_{eff}; DA15 widens day-ahead σ_p/N_{eff} broadly and tightens IDA CCGT. (Figure 7; magnitudes in the table.) Under *ID15*, the granularity instrument lands on intraday auctions: N_{eff} widens significantly in IDA for hydro, pump-storage, and wind (more granular ladders in IDA), while σ_p on IDA stays small for combined-cycle gas. The cross-market spillover lands on day-ahead: CCGT day-ahead σ_p widens by $+4.17^{***}$ EUR/MWh (anticipation-channel restructuring of DA ladders ahead of finer IDA re-trading), and hydro/pump-storage day-ahead σ_p *tightens* (negative significant). Under *DA15*, the granularity instrument lands on day-ahead: σ_p and N_{eff} widen broadly for all three dispatchable techs in DA (CCGT σ_p $+0.68^{***}$, hydro $+1.80^{***}$, pump $+0.92^{***}$; CCGT N_{eff} $+1.89^{***}$). The intraday side shows the substitution: CCGT IDA σ_p *tightens* (-0.68^{*}) — consistent with combined-cycle gas pulling its structured ladder out of IDA into DA in critical hours (cross-market substitution under DA15, consistent with the Spec B levels in (v)). Wind reads as a price-taking baseline throughout.

Behaviour change, by reform.

ID15 (intraday-auction granularity, 2025-03-19). The dispatchable fleet pushes its bid ladder *into* IDA, where the granularity instrument now lives. In-band IDA quantity grows across all three hour-classes for CCGT, hydro and pump-storage ((v)); IDA N_{eff} widens for hydro, pump-storage and wind ((i)); the cross-market spillover lands on day-ahead CCGT σ_p, restructuring the DA ladder ahead of finer IDA re-trading ((i)). Pump-storage *leaves* the now-granular IDA channel in cleared-volume terms ((iv), surviving). Both clearing prices drop on the cutover and remain below the counterfactual for the 40-day pre-blackout window, but the joint placebo-net test does not retain the level effect ((ii)).

DA15 (day-ahead granularity, 2025-10-01). The same fleet pulls its bid ladder *out of* IDA *into* DA, mirroring the ID15 pattern. In-band DA quantity grows across hour-classes for all three dispatchables; in-band IDA quantity falls ((v)). DA σ_p widens for all three dispatchables and CCGT N_{eff} widens broadly; CCGT IDA σ_p *tightens* as combined-cycle gas pulls its structured ladder out of IDA into DA ((i)). Day-ahead CCGT cleared volume scales up by $\sim$3.4 GWh/day (the only surviving level effect, (iii)). Day-ahead pump-storage cleared rises directionally but is borderline at joint 95% ((iv)).

Common mechanism. Both reforms produce the same upstream-migration of the dispatchable fleet's bidding into the market that has just become 15-min-priceable — the closed-form sequential-market model in § 3 predicts the signs and the per-tech comparative statics.

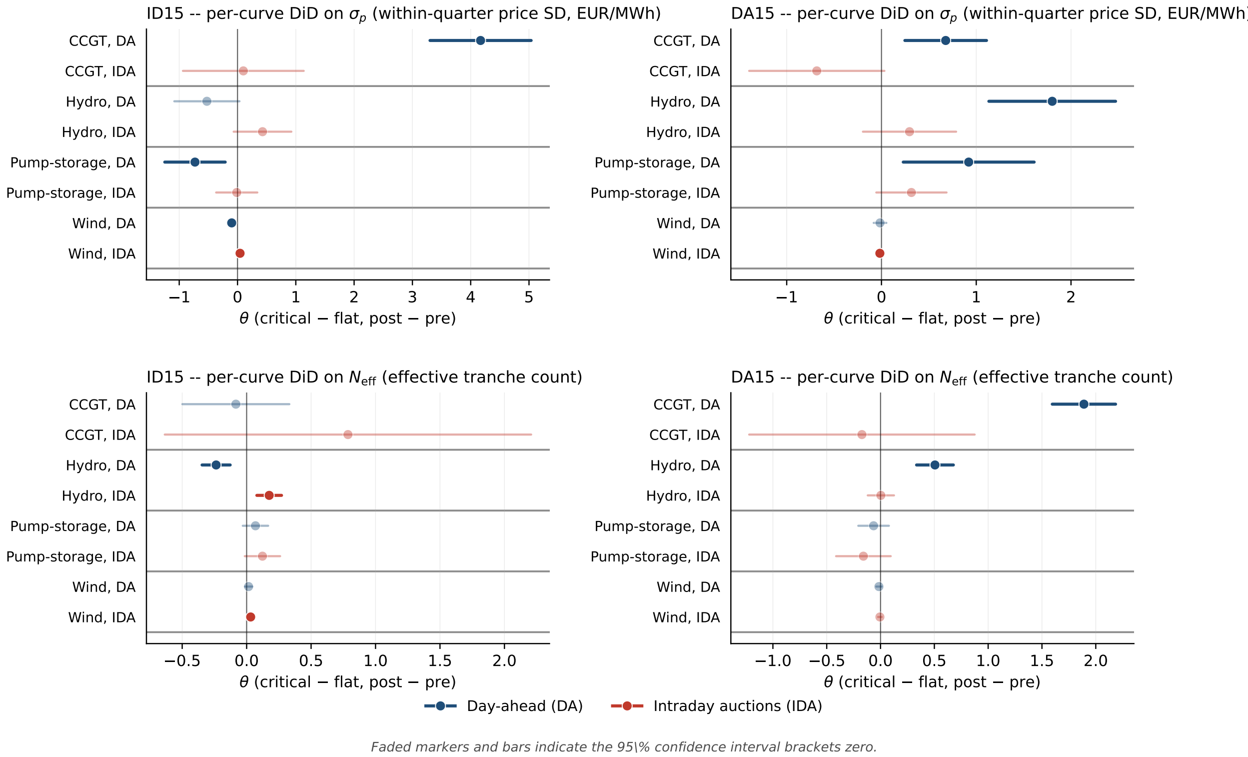

Figure 7: **Spec C per-curve DiD on bid-shape outcomes at the window-specific p90 bandwidth.** *Top row:* σ_p DiD (within-quarter price SD); *Bottom row:* N_{eff} DiD (effective tranche count). *Left column:* ID15. *Right column:* DA15. Each row inside a panel is one (tech, market) cell; blue = day-ahead, red = intraday auctions. Faded markers indicate the 95% CI brackets zero. Bandwidth h varies by (window, market): ID15 uses h=50 (DA) / h=62 (IDA); DA15 uses h=50 (DA) / h=58 (IDA). The picture is genuinely market-specific: ID15's effects concentrate in IDA N_{eff} (more granular intraday ladders) and DA CCGT σ_p (anticipation-channel widening); DA15's effects concentrate in DA across all dispatchables, with CCGT IDA σ_p tightening (substitution into DA).

5. Robustness: surviving findings hold; price effects do not

Takeaway. Bid-shape findings survive the standard threats; the BSTS DA15 combined-cycle gas cleared scale-up survives at joint 95%; price effects do not. The single fragile bid-shape outcome at the p_{90} baseline is DA15 day-ahead *pump-storage* σ_p (pre-trend placebo $+1.13^{***}$ exceeds the headline DiD $+0.92^{***}$, § 5(b)); CCGT σ_p and N_{eff} are robust (the CCGT σ_p pre-trend runs opposite the headline, so de-trending strengthens it).

Pre-trend placebo — the headline check. A pre-only midpoint placebo splits each reform's pre-window at its midpoint and runs a fake critical-flat DiD on the same outcome. The intuition: if a placebo on the pre-window already finds a critical-flat shift, then what the real DiD picks up may be extrapolation of a pre-existing trend rather than a reform effect. *ID15 IDA* (the relevant cell for ID15): all dispatchable σ_p placebos bracket zero — the ID15 bid-shape findings rest on a clean pre-trend.

DA15 DA: hydro σ_p placebo is clean (-0.05, ns), and CCGT σ_p placebo runs *opposite* the headline (-0.70^{***} vs. headline $+0.68^{***}$ in § 4.C), so de-trending strengthens the result. Pump-storage σ_p is the one wounded outcome: placebo $+1.13^{***}$ exceeds the headline DiD $+0.92^{***}$, so we read the DA15 pump-storage σ_p result as not separable from a pre-existing trend.

Hour-class robustness. A midday-vs-flat falsification gives sign-stable but smaller-magnitude DiDs at the p_{90} baseline: the reform reshapes the entire non-flat zone, not just critical hours — consistent with the per-hour-class BSTS picture (§ 4.B). The cleaner critical-vs-flat contrast remains the headline because flat hours are the unambiguous within-day control (no within-hour residual-demand variation to discriminate on).

Cross-border and reforzada — the two natural confounds. The cross-border-timing channel (SIDC/XBID also moved to 15-min on the same date as ID15) is a *sibling* treatment on the same reform, not an omitted confound: adding hourly cross-border net flow as a per-curve control on ID15 IDA σ_p moves the per-tech DiDs within $\pm 15\%$ (hydro $+11\%$ attenuation, wind $+12\%$, CCGT -13% expansion). The reforzada regulatory stance, present in both arms of DA15, is also not an OVB: it acts on the *level* of Fase I up-redispatch ($+60\%$ in flat overnight hours, $p < 0.001$) but not on its *within-hour SD* (-14%, $p = 0.08$). The interpretation is mechanical: reforzada parks capacity in the scarcity tier and recalls it through restriction offers (§ 7), while MTU15 reshapes the competing tier within the band — two non-overlapping mechanisms.

Per-CCGT-firm decomposition of the DA15 σ_p widening. The pooled DA15 CCGT DA σ_p of $+0.68^{***}$ is driven by Naturgy and Iberdrola: GN $+2.58^{***}$ (se 0.49), IB $+0.89^{**}$ (se 0.43); Endesa and EDP-HC do not contribute. The Naturgy effect is not a flat-baseline artefact at the p_{90} bandwidth: σ_p rises in both arms but more in critical hours.

Bandwidth h. Why p_{90}: it isolates the competing tier of CCGT bids (typically EUR 100–110/MWh against MCP EUR 60–90) and excludes the parked tier above the ceiling, which is a Fase I-recall placeholder rather than a strategic price (§ 7). Including the parked tier would bias σ_p upward by a measurement-driven cross-tier spread instead of a behavioural signal. The qualitative picture is robust to ± 10 EUR/MWh shifts around p_{90}. At wider bandwidths $h \in \{100, 140, 200\}$, CCGT σ_p magnitudes inflate 5–10$\times$ and several DiD coefficients flip sign because the cross-tier spread itself moves across the cutover for reasons unrelated to within-tier bidding — exactly the artefact p_{90} avoids.

Pump-storage seasonal control (Spec A). The DA15 day-ahead pump-storage rise is borderline at joint 95% in the main BSTS because of cross-season baseline drift between summer pre and autumn post. A same-calendar DiD restricted to Oct–Dec 2024 vs. Oct–Dec 2025 gives $+237^{***}$ MWh per critical clock-hour ($t = 5.0$) and a Fourier deseasonalisation gives $+350$ ($t = 7.4$) — both retain the directional finding when the demand-mix story is held fixed.

6. Continuous-market response at the three structural reforms

Takeaway. The continuous (XBID) market's structural response is concentrated at ID15: trade count surges +190% placebo-net at the cutover — the mechanical effect of the contract clock-grid moving from 60-min to 15-min in delivery. Total energy turnover *falls* (smaller average trade), and the volume-weighted mean trade price drops by $\sim$ 66 EUR/MWh placebo-net, mirroring the auction-clearing-price drop in § 4(ii). DA15 produces a smaller secondary effect (+30% daily GWh placebo-net, borderline trade-count rise, no price effect). ISP15 (the December 2024 settlement-period reform) leaves continuous-market volumes and prices unchanged — consistent with its scope being settlement, not contract structure.

Specification. BSTS-style counterfactual on daily ES-leg continuous-market series (one of `seller_zone` or `buyer_zone` = `10YES-REE------0`), with wind+solar+gas covariates and a weekly seasonal cycle (same setup as § 3.A). *Units.* All quantity outcomes are *energy* (MWh = MW $\times$ delivery hours = `quantity_mw` $\times$ `mtu_minutes`/60, summed per day), not raw MW sums: cross-reform comparability requires the stock (energy delivered), not the rate, since post-MTU15 each clock-hour holds four 15-min contracts where pre-MTU15 it held one 60-min contract. The volume-weighted mean price is energy-weighted (weight = MWh of each trade). Three reform cutovers: *ISP15* (settlement period 60 → 15 min, 2024-12-11; pre 2024-06-14 → 2024-12-10 holds the IDA reform constant, post 2024-12-11 → 2025-03-18 stops at ID15); *ID15* (intraday auctions + continuous market → MTU15, 2025-03-19; pre 2024-12-11 → 2025-03-18 post-ISP15, post 2025-03-19 → 2025-04-27 pre-blackout); *DA15* (day-ahead → MTU15, 2025-10-01; pre 2025-04-28 → 2025-09-30 post-blackout, post 2025-10-01 → 2025-12-31). Each real arm has a 2023/2024 same-calendar placebo with a fake cutover.

Reform	Outcome	Real	Placebo	Placebo-net
ISP15	n_{trades}/day	$-3{,}120^{***}$	$-1{,}231^{***}$	$-1{,}889$ (brackets 0)
	GWh/day	-3.85^{*}	-2.23^{**}	-1.62 (brackets 0)
	VW-price (EUR/MWh)	-15.1 ns	-27.4^{***}	$+12.3$ (brackets 0)
ID15	n_{trades}/day	$+22{,}335^{***}$	$+1{,}381^{*}$	$+20{,}954^{***}$
	GWh/day	-7.19^{***}	$+6.72^{***}$	-13.91^{***}
	VW-price (EUR/MWh)	-80.1^{***}	-14.1^{***}	-66.0^{***}
DA15	n_{trades}/day	$+5{,}018^{**}$	$+257$ ns	$+4{,}761$ (borderline)
	GWh/day	$+3.85^{***}$	-1.35 ns	$+5.20^{**}$
	VW-price (EUR/MWh)	$+4.7$ ns	-4.7 ns	$+9.4$ ns

Notes: BSTS posterior means; tail-probability stars $^{*}p<0.10$, $^{**}p<0.05$, $^{***}p<0.01$, ns = posterior interval brackets zero. Placebo-net joint significance based on independent posteriors propagating two variances.

(vi) ISP15: settlement scope, no contract-structure response. The December 2024 reform changed how REE settles imbalance positions (60→15-min ISP), but it did not change the OMIE continuous-market contract grid: traders still bought and sold hourly delivery contracts before and after. A null effect is therefore the correct prediction. Trade count drops 22% raw and energy turnover drops 14% raw, but the same drift is in the 2023 placebo (settlement bookkeeping innovations are not the source) and all three placebo-net intervals bracket zero. We report this as a methodological sanity check: a settlement-only reform shows nothing on contract-structure outcomes, consistent with our specification's reading of where the reform lives.

(vii) ID15: the contract grid moves from 60 to 15 minutes — the structural reform. Pre-reform, each clock-hour of delivery hosted a single 60-min continuous-market contract; post-reform, four separate 15-min contracts. Three of our four outcomes pop accordingly:

- *Trade count.* Each hour now offers four matching opportunities where there used to be one, so the trade-count surge is mechanical at first order: a raw +208%, placebo-net +190% (+21,000 matches/day, joint 95% well clear of zero). Even "mechanical" is informative — it confirms market participants *used* the new finer grid rather than continuing to trade only on the hourly aggregate.
- *Energy turnover.* The placebo-net is *negative*, −13.9 GWh/day: more contracts, but each smaller, and the total energy changing hands drops. Behavioural read: the granularity reform did not pull additional energy into the continuous market; it sliced the same (in fact a little less) energy into smaller pieces. The drop is partly because some hourly counterparties drop out when they don't see a single 60-min match on offer, and partly because dispatchables migrated bidding into the now-finer IDA auctions instead (§ 4(v)).
- *VW-price.* The continuous market clears in equilibrium with the auctions, and we see the same ID15 day-of-cutover clearing-price drop here as on auction-cleared prices (−66 EUR/MWh placebo-net, joint 95% clear of zero) — the cross-market spillover documented for ID15 in § 4(ii).

(viii) DA15: a smaller, indirect continuous-market effect. Day-ahead granularity does not change the continuous-market contract grid (already 15-min post-ID15); only the day-ahead *reference price* that continuous traders watch becomes finer. With a finer DA reference, traders reposition more aggressively in continuous: +5.2 GWh/day placebo-net (+30%, joint 95% retains), +4,800 trades/day placebo-net (borderline at joint 95%). The price effect is statistical noise (+9 EUR/MWh, brackets zero). The interpretation is that DA15's continuous-market footprint is incremental and proportional to how much tighter the day-ahead reference now is, not a structural break — exactly as expected from a reform that touches the reference market but not the continuous-market contract architecture.

7. Two market-structure facts that frame the bidding evidence

The two-tier CCGT bid curve and the day-ahead/intraday contrast. Every Big-4 CCGT fleet runs a two-tier day-ahead offer: a low *competing* tier near marginal cost (∼EUR 100–110/MWh) and a high *parked* tier above the ∼EUR 200 clearing-price ceiling — shares above the ceiling: Naturgy 75%, Endesa 22%, Iberdrola 22%, EDP-HC 19%. Withholding is firm-wide, not a Naturgy story. The same fleets bid a single competitive tier at EUR 82–96/MWh with ∼0% withheld in the intraday auction, after Fase I has cleared. A unit's marginal cost does not change between two auctions hours apart. *The parked tier is a placeholder, not a Fase I bid:* Fase I up-redispatch is settled through a *separate restriction offer* submitted to REE (procedure PO 3.2, settlement PO 14.4 ap. 20.1; see also CNMC, 2025b,a), with its own bid prices that are unrelated to the day-ahead parked-tier numerical value. Earlier drafts of this memo conflated the two; the correction matters because it justifies measuring bid shape and bid levels in the *competing tier only* — the bandwidth choice in § 1 (window-specific p90, $h \approx 45$–62 EUR/MWh) puts the parked tier outside the in-band region and isolates the strategic action. MTU15-DA grants CCGT a within-hour degree of freedom over its competing tier; the +59 GWh/day CCGT auction-cleared scale-up in § 4(iii) is the competing tier expanding into the four quarters of each peak hour.

Market concentration migrated from wholesale to ancillary services. Wholesale-market HHI hovers around 1 100 over 2016–2024 (below the moderate-concentration threshold of 1 500) and falls mid-day when solar peaks (solar is less concentrated than the Big-3). Day-ahead restrictions upward HHI above 4 000; deviations upward above 6 000; tertiary upward around 5 000. Median of 4 firms with accepted upward restrictions bids. As renewables eroded wholesale-market power, technical and locational requirements concentrated the ancillary markets that absorb their variability.

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